

Auditory setup: parts and budget

Prepared for discussion with Dr. Luan

October 9, 2026. For planned auditory recordings in sedated sheep. All prices are in United States dollars. Blue product names link to sellers or supplier quote pages. All options reuse a compatible lab computer and our neural recorder; their connections remain to be confirmed.

Proposed next step

- Confirm the required frequency range and the timing inputs available on our neural recorder.
- If tones below 20 kilohertz are sufficient, start with the basic option below. For the paper's 1.4–32 kilohertz range, obtain quotes for the compact proposal on page 4 and check what the lab already owns.

1. Basic setup: selected tones below 20 kilohertz

Computer → sound interface → powered speaker

Measurement microphone → sound interface; timing reference → neural recorder

Parts and price

Part and purpose	Number	Source / condition	Line cost
Behringer UMC204HD sound interface	1	eBay / new	97.98
JBL 305P MkII powered speaker	1	Sweetwater / new	99.00
Dayton EMM-6 measurement microphone	1	eBay / new	66.93
Hosa HMIC-010 microphone cable	1	Sweetwater / new	17.99
Hosa HSS-005 audio cable, \$14.95 each	2	Sweetwater / new	29.90
Gator RI-MICTP-FBM microphone stand	1	Sweetwater / new	34.99
Priced parts; borrowed calibrator			346.79
Optional: REED R8090 sound-level calibrator	1	DigiKey / new	219.00
Priced parts; purchased calibrator			565.79

Budget

- **Borrowed calibrator:** \$346.79 parts + \$80 estimated delivery/tax reserve = **\$426.79 working allowance.**
- **Purchased calibrator:** \$565.79 parts + \$100 estimated reserve = **\$665.79 working allowance.** The recorder-specific timing connection is additional in either case.

Timing and sound calibration

- **Actual alignment and drift: unmeasured.** Computer buffering adds delay; one 256-sample buffer at 48,000 samples per second is 5.3 milliseconds. This is an illustration, not a measurement of this interface.
- Record repeated timing references with the neural signals and use the microphone near the ear to measure sound arrival. Air adds about three milliseconds per metre; the speaker and converters add further delay.
- Check the sound level of each tone at the ear. Recording the microphone only on the sound interface still requires alignment with the neural timeline.

Decisions for the meeting

- Confirm the frequency range, recorder timing inputs and parts already available in the lab.
- Set a spending limit for supplier quotes and choose which option to develop first.

The speaker's \$99 price is a limited-time sale. Delivery/tax reserves are estimates. Sources: [computer buffering](#); [sound travel](#); [delay estimates](#). A sound output is not automatically compatible with a digital timing input.

2. SONIC paper sound chain

\$7,357.58 priced subtotal + supplier quotes; measurement kit included.

Purpose

- Follow the paper's sound-delivery hardware for its **1.4–32 kilohertz** tones: sound generation device → SA1 amplifier → MF1 speaker.
- Include the paper's satellite-based time reference. The paper's Connexus neural recorder is excluded; we would use our own system.

Parts and price

One output channel. Starred items are our reproduction accessories, not exact accessories identified by the paper. National Instruments makes the USB-6361; Tucker-Davis Technologies supplies the speaker and amplifier.

Part and purpose	Number	Source / condition	Line cost
USB-6361 sound generation/recording, 782255-01	1	Newark / new	3,744.42
TimeMachines TM2500C time reference	1	New / backorder	799.99
MF1-M speaker kit	1	Supplier quote	Quote
SA1 speaker amplifier	1	Supplier quote	Quote
ZB1PS powered enclosure*	1	Supplier quote	Quote
781513-01 power supply*	1	Newark / new	123.42
780534-01 locking computer cable*	1	DigiKey / new	69.75
763830-01 United States power cord*	1	Newark / new	51.00
781233-02 cable noise filter*	1	DigiKey / new	38.25
Hosa BNC-58-103 output lead*	1	Pro Audio / new	13.53
Priced sound parts			4,840.36
Proposed measurement kit (page 4)	1	New amplifier	2,517.22
Priced subtotal before quotes			7,357.58

Budget and remaining items

- **Supplier quotes required:** MF1-M speaker, SA1 amplifier and ZB1PS powered enclosure. Confirm speaker kit contents and included power/cable accessories.
- **Additional:** microphone connection, recorder timing connection, network wiring, delivery/tax and labor. The time reference is currently backordered.

Timing and sound calibration

- **Paper result: 56 milliseconds of neural-processing delay.** This covers neural measurement, filtering and collecting data for decoding. It excludes sensory processing and computer calculation time; it is not a sound-to-neural timestamp error.
- **Alignment and drift:** the paper does not quantify remaining sound-to-neural timing error or long-session drift. Its time reference specifies below one microsecond plus network variation; this is not sound-arrival accuracy.
- Buying these sound parts does not reproduce the paper's neural-processing delay. Test our recorder alignment and at-ear sound level separately.

Decision needed

Is close reproduction of the paper's sound hardware worth the added time-reference cost?

Sources: [SONIC version 1](#), [pages 5, 7–9 and 11](#); [sound-device specifications](#); [time reference](#); [speaker](#). The separate 11-millisecond result was a one-session analysis. Typical free-running clock scale at 25 degrees Celsius: 180 milliseconds per hour versus ideal time, not recorder-relative drift. SONIC used awake sheep; Rice's sedated-sheep performance is unverified.

Market alternative: [seller-new 782255-01](#) for \$3,498.95 saves \$245.47; confirm accessories and condition.

3. McGinley supplied setup

\$10,068.97 priced subtotal + supplier quotes; measurement kit included.

Purpose

- Follow the supplied slides: compatible desktop computer → equipment chassis → ED1 driver → ES1 speaker.
- Retain the slides' audio and signal input/output modules. Our neural recorder replaces the slides' Neuropixels equipment.

Parts and price

One output channel. Most chassis parts are used; seller testing does not establish current calibration. The desktop requires a compatible free expansion slot.

Part and purpose	Number	Source / condition	Line cost
PXle-1082 equipment chassis	1	eBay / used	2,650.00
PXle-6363 signal input/output board	1	eBay / used	2,495.36
PXI-4461 audio module, hybrid 186900-11L	1	eBay / used	699.99
PXle-8360 chassis connection module	1	eBay / used	493.20
PCIe-8361 computer connection card	1	eBay / seller-new	197.00
779500-03 rack cable, 3 metres	1	eBay / seller-new	89.85
SHC68-68-EPM cable, 192061C-02	1	eBay / used	89.99
BNC-2090A signal connection panel	1	eBay / used [‡]	613.77
ED1 driver + enclosure/power supply	1	eBay / used	209.06
ES1 speaker + matching cable	1	Supplier quote	Quote
Hosa BNC-58-103 output lead*	1	Pro Audio / new	13.53
Priced sound parts			7,551.75
Proposed measurement kit (page 4)	1	New amplifier	2,517.22
Priced subtotal before quotes			10,068.97

Budget and remaining items

- **Supplier quote required:** ES1 speaker with its matching high-voltage cable. The ED1 bundle already includes its powered enclosure and supply.
- **Additional:** microphone/recorder connections, any missing chassis power cord, delivery/tax and labor. Confirm the actual audio module's hybrid connector and calibration before buying.

Timing and sound calibration

- **Frequency limit:** ES1 is specified for **4–110 kilohertz**; it cannot cover the paper's 1.4–4 kilohertz tones. Audio-module flat output extends to 92.1 kilohertz at 204,800 samples per second.
- **Documented fixed delay:** the audio module's output filter adds **0.15625 milliseconds** at that rate. Speaker and air delay are additional; this is not whole-system timing accuracy.
- **Alignment, drift and at-ear sound accuracy: unmeasured.** Module clock settings, recorder connections and acoustic calibration need to be checked together.

Decision needed

Use this route only if its frequency range and chassis architecture fit our experiment, or substantial hardware is already available.

Sources: supplied diagram; supplied parts slide; audio module, pages 1, 19 and 24; speaker; hybrid variant. Internal-clock scale: 72 milliseconds per hour versus ideal time, conditional on calibration and aging; a different reference changes this figure. Recorder-relative drift is unknown. Speaker raw ± 11 -decibel variation is not calibrated at-ear accuracy.

[‡]Connection-panel seller lists no returns. Starred output cable is our addition; quotes/connections remain extra.

4. Proposed compact Rice setup

\$5,636.47 priced subtotal + supplier quotes; measurement kit included.

Purpose

- Cover the paper's **1.4–32 kilohertz** range with a computer-connected generation/recording device, SA1 amplifier and MF2-M speaker.
- Measure sound near the ear and record repeated timing references with the neural data. This proposed setup has not been built or validated.

Parts and price

Main comparison uses a new microphone amplifier. Dagger-marked items form the shared measurement kit: \$2,517.22. Line costs cover the number listed.

Part and purpose	Number	Source / condition	Line cost
USB-6361 sound generation/recording, 781442-01	1	Newark / new	3,030.42
MF2-M speaker kit	1	Supplier quote	Quote
SA1 speaker amplifier	1	Supplier quote	Quote
ZB1PS powered enclosure	1	Supplier quote	Quote
Earthworks M50 measurement microphone [†]	1	eBay / new	1,299.00
Sound Devices MM-1 microphone amplifier [†]	1	B&H / new	945.00
REED R8090 sound-level calibrator [†]	1	DigiKey / new	219.00
Hosa HMIC-010 microphone cable [†]	1	Sweetwater / new	17.99
Gator RI-MICTP-FBM microphone stand [†]	1	Sweetwater / new	34.99
Panasonic LR6XWA/2SB battery, \$0.62 each [†]	2	DigiKey / new	1.24
Pomona EM4970-36# output lead	1	DigiKey / new	23.68
Hosa PHX-106F-BULK microphone connection lead	1	B&H / new	13.95
Yageo 100,000-ohm resistor, \$0.10 each	2	DigiKey / new	0.20
763830-01 United States power cord	1	Newark / new	51.00
Priced subtotal before quotes			5,636.47

Budget and remaining items

- **Supplier quotes required:** MF2-M speaker kit, SA1 amplifier and ZB1PS powered enclosure. Recorder connection, delivery/tax and labor are additional.
- **Used alternative:** a **\$604.50 B&H MM-1** reduces this subtotal to **\$5,295.97**; listed moderate wear, 90-day seller warranty. The shared kit becomes \$2,176.72.

Timing and sound calibration

- **Frequency coverage:** speaker nominally 1–80 kilohertz; microphone to 50 kilohertz. The calibrator's **0.5-decibel specification applies at one kilohertz only**; full-band at-ear uncertainty remains unmeasured.
- **Candidate alignment goal:** sound arrival and neural timestamps within **one millisecond after correction**. This is a goal for discussion and testing, not achieved performance or SONIC's processing-delay measure.
- Check repeated timing references, microphone arrival, converter channel offsets and adapted microphone wiring. Actual residual alignment and drift are unknown.

Decision needed

Obtain this route's speaker/amplifier quotes if the full paper frequency range is required.

Device power/cable and speaker stand/cable are included. Microphone lead and bias returns require wiring review. Arduino or a field-programmable gate array still needs analog/acoustic hardware and development.

Sources: speaker; microphone; microphone amplifier; calibrator; sound device. Clock scale: 180 milliseconds per hour versus ideal time, typical at 25 degrees Celsius; actual recorder drift is unknown. Detailed comparison.